Forecasting fuel retail sales volume using machine learning for sustainable decision-making

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Abstract. Purpose: Forecasting fuel retail sales volume is important for sustainable decision-making and process management. Prediction of the petrol products sales volume influences the planning of the effective distribution of fuels. Therefore, it impacts all other supply chains and each country's economy. Method: This research uses machine learning to forecast fuel retail sales volume. The choice of this method and the research subject are based on the complexity of the decision-making process in energy carriers' sales. Machine learning model (LightGBM) and two other state-of-the-art models (ARIMA, Holt-Winters) were compared. Performed calculations are based on real-world data from single fuel company distributors. Results: The three machine learning models are proposed and examined to address this research gap. There is a baseline model proposed along with state-of-the-art models and the results are based on comparative analysis of MSE and RMSE. Finally, a real-world sales data set from a fuel dispenser is analyzed to determine the veracity of the model argument. Conclusion: The best results were obtained by the LightGBM model, utilizing both future covariates and forecasted fuel prices. This model demonstrated the best performance according to all metrics. Performed experiments proved that the best model can support strategic reasoning in organizations interested in forecasting fuel sales. The use of machine learning to forecast retail fuel sales volume for sustainable decision-making is a good example of leveraging modern technology to optimize business processes and make more informed decisions.

Keywords: Sales, Prediction, LightGBM model, Randomforest, Fuel

1 Introduction

Balanced and sustainable decision-making is often opposed to temporary or day-to-day management processes in the unstable business environment to present its characteristics [1]. Unlike ongoing and day-to-day decision-making, sustainable decision-making requires a reliable plan, strategy and models to forecast its outcomes [2]. Sustainable decision-making aims to overcome business instability and offers businesses a sustainable competitive advantage relying on the multi-perspective models [1]. Although sustainable decision-making does not exclude the environmental component or sustainability concept, it focuses more on the longitudinal approach, strategic approach and

mechanism design for multi-perspective models. In this context, the energy and fuel sector are an example of the most challenging sector of the economy. There are environmental issues related to the energy sector transformation and political tensions revolving around energy security, altogether with social expectations related to the fuels' accessibility [3]. Therefore, sustainable decision-making regarding the retail sale of fuels requires processing large amounts of data, covering different variables influencing the fuels' market. One of the most significant problems in this area is fuel retail sales volume forecasting [4], and usually, this is addressed in the scientific literature by combination of classic statistical and machine learning methods. However, the outcomes of those forecasting procedure types depend on the subject of analysis and its accuracy. The forecasting procedures accuracy is measured by MSE (Mean Square Error) and MAPE (Mean Absolute Percentage Error) to check fuel retail sales volume prediction.

Accurate forecasting allows for more sustainable supply chain management and sustainable fuel stocks decision-making. Thanks to the fuels retail sales forecasting, it is possible to reduce the Forrester effect, which is unexpected or unpredicted demand distortion that travels upstream in the supply chain [5]. Scientific and business literature combining sustainable decision-making in fuel retail sales volume are also scarce. However, the existing researches are primarily concerned with collectively forecasting volumes and prices in the fuel market. The classical relation between fuel price and retail sales volume has rarely been addressed in scientific publications. However, among those publications, classic statistical and machine learning methods are used. Even though the fuel market plays a pivotal role in the modern economy [6], only a few scientific studies forecast retail fuel sales volume.

This research aims to develop approaches for forecasting the volume of retail sales of fuels. The main contributions of this research are as follows: (I) developing classical statistical approaches for forecasting retail sales of fuels, (II) developing machine learning approaches for forecasting retail sales of fuels, (III) comparing and analyzing developed approaches, and (IV) extending the existing literature on the subject. These scientific research results can inspire other researchers and business practitioners looking for the experiments' results and want to improve their fuel retail sales volume forecasting models.

This paper is organized into five sections. Besides this introduction, the scientific background is discussed in the second section. Next the pre-processing of the input data and methods are presented. The next part of the paper presents the results of the research experiments. The last sections are discussion and conclusions, addressing possible future research directions.

2 Literature background

Sustainable decision-making is a consequence of the adopted organization management style [1], organizational routines [7] and culture [8, 9]. Energy sector organizations understand that sustainability is more comprehensive and can be supported by sustainable decision-making as environmental challenges become more urgent. This creates an incentive for implementing comprehensive intelligent, data-driven decision models that

support the interactive evaluation of decision alternatives required by the business environment [10].

Sales forecasting volume can use statistical methods, historical data, and market analysis. This involves identifying trends and patterns in historical sales data and utilizing these insights to make accurate projections regarding future sales volumes, trends, and other relevant metrics [11]. Sales forecasting is a crucial element of business planning because it enables businesses to set realistic goals and make informed decisions regarding resource allocation, marketing strategies, and other crucial aspects of their operations. Incorporating external factors such as economic indicators, competitor analysis, and customer behavior can improve sales forecast accuracy. This is a canonical problem with a fairly classic and well-established spectrum of methods, which seems to have stagnated prior to the introduction of deep learning-based methods. Recently, a new machine learning technique has been implemented, which supports the explanation of cutting-edge black-box prediction models [12]. However, non-transparent black-box machine learning (ML) models such as random forests, boosting, support vector machines (SVM), and neural networks achieve significantly better predictive performance than simple, interpretative models such as decision trees, naive Bayes, or decision rules [13]. This is one explanation for the low utilization and acceptability of predictive ML models in sectors requiring transparency and clearness of choices. Dietterich [14] evaluates a number of well-established ML techniques for sequential data prediction on a variety of time series. The idea of ML techniques on sequential data expand to Corporate Bankruptcy Prediction [15] and use of Recurrent Neural Networks (RNN) [16]. Gamboa offers a recent overview of deep learning for time-series forecasting [17]. Due to their effectiveness with nonlinear and high-frequency time series, neural networks have gained popularity in regression models

The petroleum retail industry is a classic example of price and demand fluctuations caused by a variety of factors, such as shifts in global oil prices due to political unrest in oil-producing nations and currency devaluation. Retail gasoline (gasoline) stations alter gasoline prices to remain competitive [18]. When one gas station lowers its price, another gas station responds by lowering its price even further, resulting in a cycle of price reductions that continues until the profit breaks even. At this point, a gas station may "relent," after which all other gas stations follow suit, and the cycle begins again [19]. Therefore, accurate fuel station sales forecasting is essential for station owners to make well-informed business decisions.

A fuel consumption forecasting system was developed in the early 2000s to predict fuel consumption at Petronas petrol stations in Malaysia [20]. This study utilized a rule-based expert system consisting of two logical components: a knowledge base and an inference engine. The results were satisfactory, but results from the testing showed the flaws of this method. Fuzzy linear regression tree [21] was introduced in problem of forecasting sale volume of petroleum products. In the study from 2012 The sales forecasting methodology presented in this study utilizes an adaptive fuzzy modeling technique (fuzzy linear regression trees). With a provided training dataset, the algorithm progressively assesses whether a fuzzy linear regression tree, which is a weighted combination of several local linear regression models, can outperform the initial global linear model [22]. Another research from 2018 focused on forecasting petrol products

primarily investigates various approaches, using clustering and classical decision tree. However, these methods tend to identify patterns in forecasting schemes through clustering, rather than accurately predicting specific sales volumes [23].

3 Method

This method section, organized into several subchapters, comprehensively describes the different forecasting methods utilized for time-series analysis. These methods include ARIMA, Holt-Winters, and LightGBM. Each subchapter provides in-depth detail about each method's characteristics and presents specific examples of their applications in various fields. First subchapter describes the data collected from fuel distributors, the process of transforming raw data into a usable format, and the creation of a data pipeline for a machine learning model. Fuel price data as additional covariates subchapter explains the use of historical fuel prices as additional covariates in the model and the method of using percentage price differences for more effective forecasting. Fuel price differences prediction section focuses on the separate forecasting experiment conducted to predict price changes and prevent data leakage, comparing the performance of LightGBM, ARIMA, and NBeats algorithms. The final section discusses the training of the LightGBM model on historical refueling data, outlining the model's use of future covariates and forecasted fuel prices.

ARIMA is a versatile and expressive model that combines autoregressive (AR), differencing (I), and moving average (MA) components to represent time series. The AR component captures temporal dependence on past values, the MA component captures dependence on past white noise error terms, and the I component represents differencing to transform non-stationary time series into stationary ones [24]. ARIMA has been widely used in diverse fields, such as forecasting electricity demand [24] and stock prices [25–27], and can be combined with advanced methods like Deep Learning for tasks such as high-frequency financial time-series forecasting [28].

Holt-Winters models are a class of exponential smoothing methods that are widely used in time series forecasting. The Holt-Winters models are named after Charles C. Holt and Peter G. Winters, who introduced the method in their seminal paper [29]. The Holt-Winters models are a type of exponential smoothing method that uses three smoothing parameters to capture the level, trend, and seasonality of a time series. The basic idea behind exponential smoothing is to give more weight to recent observations and less weight to older observations. This method has been widely used in forecasting applications in various fields, for example, to forecast demand for products, sales of consumer goods [30], and energy consumption [31, 32].

Machine learning method utilizing Light Gradient Boosting (LightGBM), as it was proven to be effective and robust in tabular data modeling with the possibility to handle time series forecasting after proper data preprocessing. LightGBM is a gradient-boosting framework that uses a tree-based learning algorithm for classification and regression tasks, similar to XGBoost [33]. It was developed by Microsoft Research and was introduced in 2017 [34]. The main principle of LightGBM is to build decision tree leaf-wise rather than level-wise. This approach can lead to faster training times and better

accuracy than other gradient-boosting frameworks. LightGBM also uses a technique called gradient-based one-sided sampling (GOSS) to reduce the number of samples used in the training process while maintaining [35] accuracy [34]. These principles are effective in various applications, including stock price predictions [35], cryptocurrency price trend forecasting [36], anomaly (malware) detection [37], or machine-learning-assisted medical diagnosis [38]. The experiment aimed to accurately predict refueling volume in liters on a predefined period, according to mean squared error (MSE) and mean absolute percentage error (MAPE) metrics.

3.1 The data

This section describes the general meta-data knowledge discovery of company-collected data. In subsequent sections of this paper were discussed the methodologies and potential applications of data mining on time series collected from fuel distributors.

The data to be described is from one of the company's stations. Taking advantage of recent developments in information technology, this fuel distributor has modernized its business operations at the store level by connecting to PC-based servers via a specific interface, and at the corporate level by implementing highly scalable parallel processing architectures. Figure 1 describes generally the data pipeline for a machine learning model using gas distributor data involves several steps: raw data collection, data warehousing, data export, data transformation, and the machine learning model itself. These steps work together to convert raw data into meaningful insights and predictions, thus enabling better decision-making and process management.

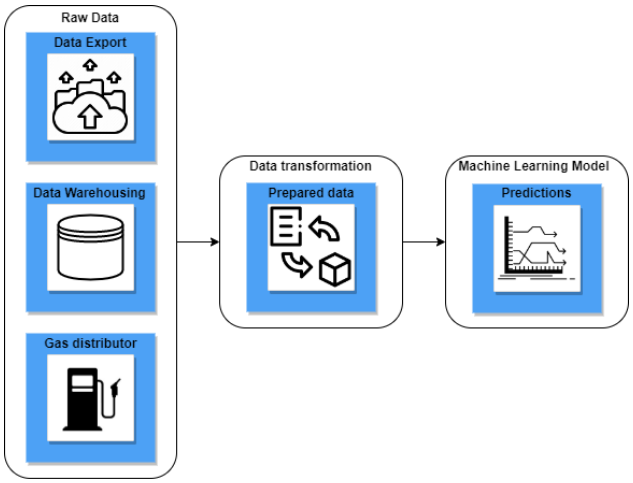

Fig. 1. The data pipeline.
Source: Author's elaboration

The data obtained from the company's fuel volume sales have not been processed and it was mandatory to curate it. Raw data consists of 492178 data points from all

distributors and type of gas. Table 1 show differences between obtained raw data and processed data to further experiment. After aggregation it was 1177 data points.

Table 1. Comparison between raw and processed data.

	Raw data	Transformed data
Data types	20 int64, 9 categorical, 4 float64	2 float 64,4 int64
Missing data	Yes* (categorical)	No
Aggregation	No	Daily sales data by gas distributor

Source: Author's elaboration

Refueling volume data has been used as the main time series during the modeling. The data spanned across the following periods with has x data points:

- Whole volume consisted of daily refueling volume from one distributor.
- Training data spanned from 2017-04-13 to 2020-03-05
- Testing data spanned from 2020-03-06 to 2020-07-01

Formally:

$$Y = \left[y_{t_1}, y_{t_2}, \dots, y_{t_T}\right] \tag{1}$$

Where, y_t is the refueling volume as the time t, and T is the maximal date in a time series. Therefore, the future-covariates matrix $X^{\{Cal\}}$ can be defined as follows:

$$X^{Cal} := \begin{bmatrix} Yr_{t_1} & M_{t_1} & D_{t_1} & W_{t_1} & DY_{t_1} & Q_{t_1} \\ Yr_{t_2} & M_{t_2} & D_{t_2} & W_{t_2} & DY_{t_2} & Q_{t_2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ Yr_T & M_T & D_T & W_T & DY_T & Q_T \end{bmatrix}, \forall Yr_t, M_t, D_t, W_t, DY_t, Q_t \in N \tag{2}$$

LightGBM algorithm requires additional predictors for handling time-series data. Therefore, the matrix of future covariates was constructed, consisting of the following indicators (for each date in a training and testing data), where:

- Year – Yr
- Month number - M
- Day of week number - D
- Number of the week in a year - W
- Number of the day in a year - DY
- Quarter number – Q

Fuel price can strongly influence the refueling volume. Therefore, the historical fuel prices per cubic meter were used. The prices originated from one of the fuel products suppliers and included the period from 2017-01-01 until the end of training data. The data originates from other source than fuel retail volume data Raw fuel prices can be challenging to forecast. Therefore, a time series of percentage price differences were used instead. Formally:

$$\Delta p_t = \frac{t\text{-}p_{t\text{-}1}}{p_{t\text{-}1}}, \Delta p_t \in R$$
$$\Delta p = [\Delta p_1, \Delta p_2, \ldots, \Delta p_T] \tag{3}$$

3.2 Fuel price differences prediction

To avoid the data leakage problem during forecasting - using fuel price differences Δp that should not be known at the prediction time - an additional meta-model was trained to predict price change, used then as covariates $X^{\Delta p}$

The figures presented below illustrate the relationship between the refueling volume prediction y and fuel price differences Δp treated as additional covariates. Figure 1 depicts a situation when values of both time series are known; the model uses the following elements for prediction:

- Past values of the primary (predicted) time series - refueling.
- Past values of covariate time series - fuel prices.

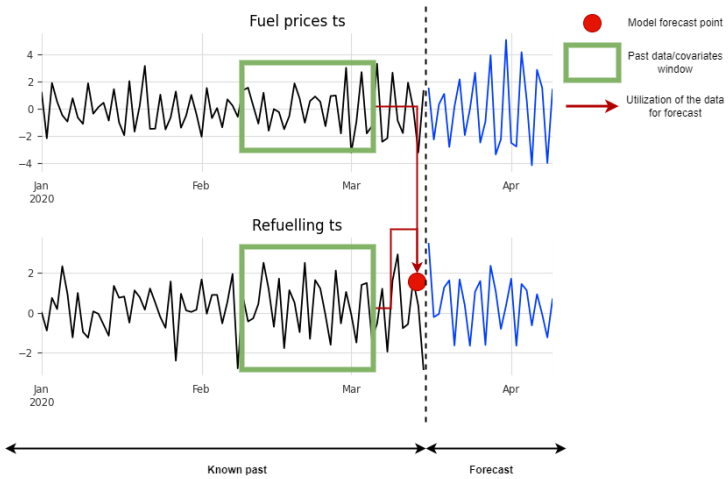

Fig. 2. Refueling time series prediction using known data.
Source: Author's elaboration

Figure 2 depicts a situation when only some of the two-time series values are known. This scenario happens when the forecast is done out-of-sample. Then the model uses its own past predictions (rolling forecasts, sometimes called 'out-of-sample') and known historical data if it fits inside the prediction window.

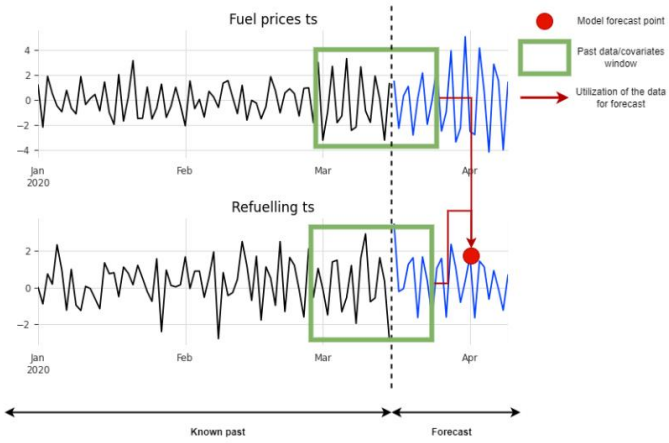

Fig. 3. Refueling time series prediction using rolling forecasts.
Source: Author's elaboration

This modeling was a separate forecasting experiment, where the following algorithms were compared in terms of MSE and MAPE metrics: LightGBM, ARIMA, and-NBeats. The fuel price difference prediction experiment included a separate training-validation loop independent from the refueling prediction. Based on the experimental results (presented in the Table 1) LightGBM was selected for forecasting.

Table 2. Fuel price differences forecasting results.

Model	MSE	MAPE
LightGBM	0.0001	1059.8537
NBeats	0.0001	1460.3736
ARIMA	0.0001	inf

Source: Author's elaboration

The large error values, in summary, are the result of very high price variability, although the inspection of LightGBM residuals indicates normality, and forecasts capture the overall price change trends, as shown in the figure below. Figure 4 on the left displays the difference between predicted and actual fuel price differences over time, offering insights into the model's prediction accuracy, where y is real data and yhat LightGBM is the LightGBM model. Figure 4 on right describes the distribution of LightGBM residuals could provide evidence of normality, indicating a good fit of the model to the data. The residuals mostly hold in range -100 to 100.

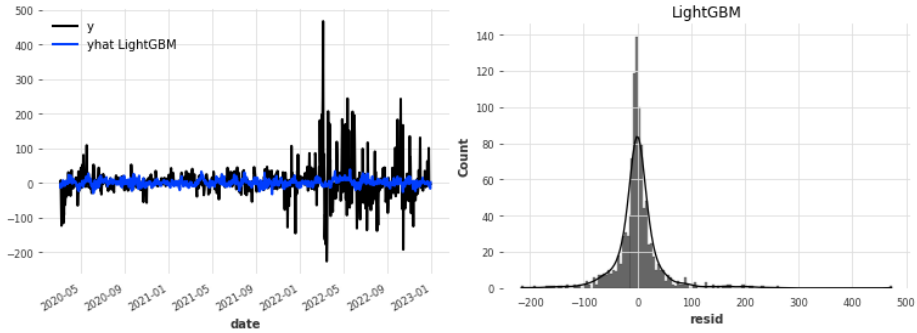

Fig. 4. LightGBM price difference prediction (left) and residuals (right).
Source: Author's elaboration

4 Results

The results of the experiment are summarized in the table 3. The best results were obtained by the LightGBM model, utilizing both future covariates and forecasted fuel prices. This model demonstrated the best performance according to all metrics (MSE and MAPE).

Table 3. Fuel price differences forecasting results

Model	MSE	MAPE
LightGBM	**88694,4968**	**17,9563**
Naive seasonal model	299658,2014	34,2128
Holt-Winters	384454,1543	41,9970
ARIMA	428295,8068	43,8615

Source: Author's elaboration

5 Conclusion

The LightGBM model developed in this research allows for achieving the lower values of MSE and MAPE than "classic" forecasting method. Therefore, the results of the research contribute to sales forecasting area by extending existing approaches about machine learning model for sales volume forecasting in fuel retail. The elaborated model can be used for improving volumes sales prediction in decision support systems performing in fuel retail. In consequence developed model allows for making more sustainable decisions by, among others, reducing the Forrester effect in the supply chain. Accurate petrol product sales forecasting enables businesses to reduce fuel inventory and improve companies' sustainable decision-making over uncertain data and models. These sales processes are affected by numerous interrelated factors, but most often considered is the classical influence of fuel price. Therefore, multi-perspective

process models are rare. The main limitation of research is using only LightGBM machine learning model. Another limitation is lack of statistical analysis between models. Another limitation is the quality (lack of data continuity) and quantity of data (small number of refueling stations) analyzed in this study. Therefore, the future research works should be related to developing also other machine learning approaches for sales volume forecasting in fuel retail, including deep neural networks. Also, deeper statistical analysis of results should be performed.

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